



NTCIP Policies

<https://ite-org.github.io/NTCIP-8001/latest/>

AASHTO / ITE / NEMA

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Front Matter

Working Group Draft

NTCIP 8001 v.2.0.0-a.4

National Transportation Communications for ITS Protocol

NTCIP Joint Standardization Policies, Processes and Procedures

 June 21, 2025

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Acknowledgements

This document was prepared through an open-source standards development process with the following active contributors:

contributors 1

Check out the full list of [contributors here](#).

In addition, the following submitted comments during the process:

- k-vaughn

The resultant document is maintained by the NTCIP Open-Source Process (OPS) Working Group (WG), a subdivision of the Joint Committee on the NTCIP. The Joint Committee on the NTCIP is organized under a Memorandum of Understanding among the American Association of State Highway and Transportation Officials (AASHTO), the Institute of Transportation Engineers (ITE), and the National Electrical Manufacturers Association (NEMA). The Joint Committee on the NTCIP consists of six representatives from each of the standards development organizations (SDOs) and provides guidance for NTCIP development.

🕒 June 21, 2025

Foreword

Overview

This document is an NTCIP Open-Source NTCIP Process, Control, and Information Management document provided as Interim for Field Release (IFR).

Open-source documents are developed using the ITS Open-Source Process, as defined in NTCIP 8008. This process provides an open standards development process that accepts issues reported by the community and resolved by peer-reviewed contributions from the community. The open source process concludes with the resultant material being approved by the defined approval process.

IFR documents are approved through a streamlined process focused on the technical experts of the community (e.g., those participating in the open-source development process) rather than through a formal ballot of industry managers.

NTCIP Process, Control, and Information Management documents define the practices and policies used by the NTCIP Joint Committee and its working groups in developing and maintaining NTCIP publications.

NTCIP 8001 NTCIP Joint Standardization Policies, Processes and Procedures v03 has been prepared for the National Transportation Communications for ITS Protocol (NTCIP) project for use by the NTCIP Joint Committee which oversees the NTCIP project, the NTCIP standard development organizations (SDOs) which jointly approve NTCIP standards, the NTCIP Coordinator who administers the project, and all other stakeholders who participate in the development and maintenance of NTCIP standards.

Approvals

IFRs are peer reviewed within the open-source process with final approval by an associated WG established by the NTCIP Joint Committee.

Approval information is provided within the online environment.

For more information about NTCIP standards, visit the NTCIP Web Site at www.ntcip.org.

User Comment Instructions

Comments can be submitted at any time. In preparation of this NTCIP standards publication, input of users and other interested parties was sought and evaluated.

Comments on open-source projects can be submitted either on the [discussions](#) or [issues](#) tab of the project.

Discussions can be initiated at any time and anyone in the community can respond, all within a public environment. Responses to discussion comments are strictly informative and may not be accurate. Discussion comments can lead to the submittal of issues that need to be resolved to clarify the standard.

Issues can be submitted at any time. Issues are triaged by the project maintainer, who will evaluate their merit, classify them (e.g., as a bug, documentation issue, omission), and in most cases respond to the submitter. Once ready, issues will be available for contributors to volunteer to address. When a volunteer has a proposed solution, it can be submitted to the project and approved in a relatively short period (when compared to the traditional standards approval process). However, updates to the projects are still version controlled so that users can reference a specific version of the project without fear of it changing.

Comments should use the templates provided on the website; otherwise they may be ignored.

History

For a history of the project, see the projects [releases](#) page.

 June 21, 2025

Introduction

NTCIP 8001 provides the framework for the development and management of the standards and other supporting documents that make up the enterprise known as the National Transportation Communications for ITS Protocol (NTCIP) joint standardization project.

NTCIP 8001 contains mandatory requirements that are always applicable and optional and conditional requirements, which may be applicable.

This document contains one informative annex.

The following keywords apply to NTCIP 8001: AASHTO, ITE, NEMA, NTCIP, process and control standard, process, policy, procedure.

This document only uses metric units.

 June 21, 2025

Section 1 General

1.1 Scope

This document specifies a set of rules for organizing, describing, and defining transportation management information to be exchanged between transportation management applications and transportation equipment. This document defines the Structure and Identification of Management Information (SMI) used in transportation-related devices. This document is applicable to the NTCIP 1200 series and other NTCIP standards that deal with device data dictionaries.

NOTE—This document relies on widely accepted conventions, generally designated as “SMIv2” and defined in the Internet Architecture Board (IAB) Standard (STD) 58.

1.2 References

The following documents are referenced by this document. At the time of publication, the editions indicated were valid.

1.2.1 Normative References

Normative references contain provisions that, through reference in this text, constitute provisions of this document. All standards are subject to revision, and parties to agreements

based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standard listed.

Identifier	Title
IAB STD 58	(RFC 2578, Structure of Management Information Version 2 (SMIv2), 1999; RFC 2579, Textual Conventions for SMIv2, 1999; RFC 2580, Conformance Statements for SMIv2, 1999)
RFC 3416	Version 2 of the Protocol Operations for the Simple Network Management Protocol (SNMP), 2002
RFC 3419	Textual Conventions for Transport Addresses, 2002
RFC 3584	Coexistence between Version 1, Version 2, and Version 3 of the Internet-standard Network Management Framework, Internet Engineering Task Force (IETF), August 2003
RFC 4001	Textual Conventions for Internet Network Addresses, 2005
RFC 4181	Guidelines for Authors and Reviewers of MIB Documents, Internet Engineering Task Force (IETF), September 2005
ITU-T X.208	Open Systems Interconnection Model and Notation – Specification of Abstract Syntax Notation One (ASN.1), 1988 ¹
ITU-T X.680 (ISO/IEC 8824-1:2015)	Information technology—Abstract Syntax Notation One (ASN.1): Specification of basic notation
ITU-T X.690 (ISO/IEC 8825-1)	Information technology—ASN.1 encoding rules: Specification of Basic Encoding Rules (BER), Canonical Encoding Rules (CER), and Distinguished Encoding Rules (DER)
ITU-T X.696 (ISO/IEC 8825-7)	Information technology—ASN.1 encoding rules: Specification of Octet Encoding Rules (OER)
BIPM SI	The International System of Units (SI) – 9th edition (2019), https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf

1.2.2 Other References

Other references are included to provide a more complete understanding of this document and its relationship to other documents.

1.2.2.1 Other Resources for Contributors

This document standardizes and tailors certain aspects of the information contained in open-sourced; however, it is not a complete replacement of that material. If you wish to learn more about open-source development, the following materials may be of interest:

Identifier	Title
IAB STD 8	(RFC 0854: 1983, <i>Telnet Protocol Specification</i> , and RFC 0855: 1983, <i>Telnet Options Specifications</i> ; J. Postel, J. Reynolds)
RFC 2021	<i>Remote Network Monitoring Management Information Base Version 2 using SMIv2</i> , 1997
RFC 2287	<i>Definitions of System-level Managed Objects for Applications</i> , 1998
RFC 2856	<i>Textual Conventions for Additional High Capacity Data Types</i> , 2000
RFC 2981	<i>Event MIB</i> , 2000
RFC 2982	<i>Distributed Management Expression MIB</i> , 2000
RFC 3014	<i>Notification Log MIB</i> , 2000
RFC 3231	<i>Definitions of Managed Objects for Scheduling Management Operations</i> , 2002
RFC 3411	<i>An Architecture for Describing Simple Network Management Protocol (SNMP) Management Frameworks</i> , 2002
RFC 3433	<i>Entity Sensor Management Information Base</i> , 2002
RFC 3877	<i>Alarm Management Information Base (MIB)</i> , 2001
RFC 4268	<i>Entity State MIB</i> , 2005
RFC 6933	<i>Entity MIB (Version 4)</i> , 2013
AASHTO / ITE / NEMA NTCIP 1201 v03	<i>Global Object (GO) Definitions—version 03</i> , published March 2011
AASHTO / ITE / NEMA NTCIP 1202 v03	<i>Object Definitions for Actuated Signal Controllers (ASC) Interface—version 03A</i> , published May 2019
AASHTO / ITE / NEMA NTCIP 8005 v02	<i>Procedures for Creating Management Information Base (MIB) Files</i> , Proposed
ISO 26048-1:2025 ²	<i>Intelligent transport systems – Roadside modules <u>SNMP</u> data interface – Part 1: Global Objects</i>
ISO/IEC 9834-1:2012	<i>Information technology – Procedures for the operation of object identifier registration authorities: General procedures and top arcs of the international object identifier tree – Part 1:</i>
ISO/TS 20684-2:2021	<i>Intelligent transport systems – Roadside modules <u>SNMP</u> data interface – Part 2: Generalized field device basic management</i>
ISO/WD 20684-3	<i>Intelligent transport systems – Roadside modules <u>SNMP</u> data interface – Part 3: Triggers</i>
ISO/WD 20684-4	<i>Intelligent transport systems – Roadside modules <u>SNMP</u> data interface – Part 4: Notifications</i>
ISO/WD 20684-5	<i>Intelligent transport systems – Roadside modules <u>SNMP</u> data interface – Part 5: Logs</i>
ITU-T X.660 (07/2011)	<i>Identical to ISO/IEC 9834-1:2012. Recommended status.</i>
ISO/WD 20684-6	<i>Intelligent transport systems – Roadside modules <u>SNMP</u> data interface – Part 6: Commands</i>
ISO/WD 20684-7	<i>Intelligent transport systems – Roadside modules <u>SNMP</u> data interface – Part 7: Support Features</i>
NEMA TS_2-2016	<i>Traffic Controller Assemblies with NTCIP Requirements</i> , 2016
	<i>IETF</i> , Generic and Common Textual Conventions (TCs), https://trac.ietf.org/trac/ops/wiki/mib-common-tcs , accessed June 10, 2021
	Perkins, D; McGinnis, E., <i>Understanding <u>SNMP</u> MIBs</i> , Prentice-Hall, Inc., 1997, ISBN 0-13-437708-7
	Stallings, William, <i><u>SNMP</u>, <u>SNMPv2</u>, and CMIP: The Practical Guide to Network-Management Standards</i> , Massachusetts, Addison-Wesley Publishing Company, 1993, ISBN 0-201-63331-0

Identifier	Title
	Larmouth, John, ASN.1 Complete , Academic Press, a Harcourt Science and Technology Company, May 1999, ISBN 0-12233-435-3, http://www.oss.com/asn1/booksintro.html (October 9, 2000)
	Dubuisson, Olivier, ASN.1 Communication between Heterogeneous Systems , June 5, 2000, ISBN 0-12-6333361-0, http://www.oss.com/asn1/bookintro.html (October 9, 2000)
	Booch, Grady, Rumbaugh, James, Jacobson, Ivar, Unified Modeling Language User Guide , September 30, 1998, ISBN 0-20157-168-4
	UML basics: An introduction to the Unified Modeling Language , www-106.ibm.com/developerworks/rational/library/769.html#N10090 .

1.2.3 Contact Information

1.2.3.1 Internet Documents

Obtain Request for Comment (RFC) electronic documents from several repositories online at:

www.rfc-editor.org

www.rfc-editor.org/repositories.html

1.2.3.2 Architecture Reference for Cooperative and Intelligent Transportation (ARC-IT)

The Architecture Reference for Cooperative and Intelligent Transportation (ARC-IT) may be viewed online at:

<http://arc-it.net>

1.2.3.3 ISO, IEC, and ISO/IEC Standards

ISO, IEC, and ISO/IEC standards can be purchased on-line in electronic format or printed copy from:

Techstreet
6300 Interfirst Dr.
Ann Arbor, MI 48108
(800) 699-9277
www.techstreet.com

1.2.3.4 IEEE Standards

IEEE standards can be purchased on-line in electronic format or printed copy from:

Techstreet
6300 Interfirst Dr.
Ann Arbor, MI 48108
(800) 699-9277
www.techstreet.com

1.2.3.5 NTCIP Standards

Copies of NTCIP standards may be obtained from:

NTCIP Coordinator
National Electrical Manufacturers Association
1300 N.17th Street, Suite 900
Rosslyn, Virginia 22209-3801
www.ntcip.org
e-mail: ntcip@nema.org

Draft amendments, which are under discussion by the relevant NTCIP Working Group, and amendments recommended by the NTCIP Joint Committee are available.

1.3 General Statements

The remainder of this document is broken into the following chapters:

- **Background:** Describes the purpose of this document
- **Object Identification:** Discusses the object identifier tree and defines the nodes related to the NTCIP effort.
- **Rules for Module Development:** Defines the rules for developing MIB modules within the NTCIP standards efforts, including the specialized fields used by the NTCIP.
- **Requirements for Agent Implementations:** Requirements for implementations of agents that claim conformance to NTCIP standards.
- **Guidelines for Agencies:** Advice for agencies on operating NTCIP devices.
- **NTCIP Structure of Management Information:** The formal MIB module that defines the structure of the NTCIP portion of the object identifier tree and key textual conventions.
- **Conversion from SMIv1 to SMIv2:** Rules for converting SNMPv1 MIBs to support SNMPv3.
- **Macro Examples:** Examples of applying the rules defined in this document.
- **History of Changes:** A list of changes made with the current version from the last major version.

1.4 Terms

For the purposes of this document, the following terms and definitions apply. Terms not defined here are used in accordance with their definitions in ISO/IEC/IEEE 24765. Words not defined

here or in ISO/IEC/IEEE 24765 are used in accordance with their definitions in *Webster's New Collegiate Dictionary*.

Term	Definition
agent	The entity that receives requests and transmits responses to the received requests.
block object type	An <u>object type</u> that represents a standardized data structure of elemental <u>object types</u> .
compatibility	The ability of two or more systems or components to exchange information. [ISO/IEC/IEEE 24765]
configurable object	An <u>object type</u> that represents a data structure that can be configured at run-time to include elemental <u>object types</u> .
context	An instance of a management information base (MIB). NOTE—A single SNMP <u>agent</u> (e.g., a unique UDP/IP address) typically represents a single <u>context</u> , but might support multiple contexts if it, for example, serves as the interface for multiple field devices (e.g., two dynamic message signs controlled by a single controller) or serves as a proxy <u>agent</u> for multiple field devices. When there are multiple contexts, there might be multiple <u>object</u> instances with the same <u>object</u> identifier and the <u>context</u> is used to disambiguate.
current	Reflecting the conditions at the present time (or at the time at which the data is time stamped) as determined by the Controller. Within the <u>context</u> of a status value, indicates the most up-to-date design of the element (e.g., <u>user need</u> , <u>requirement</u> , <u>object</u>).
data	Information before it is interpreted.
datagram	A self-contained unit of data transmitted independently of other datagrams.
deprecated	A status value that indicates the <u>user need</u> , <u>requirement</u> , <u>dialog</u> , or <u>object</u> is valid in limited circumstances, but has been replaced by another. NOTE—This definition is modified from “Understanding SNMP MIBS.” Procurements can require support for deprecated objects to enable multi-version <u>interoperability</u> (e.g., <u>backward compatibility</u>) with legacy implementations.
file	A grouping of data into a single sequence of bytes that can be referred to by <u>file</u> operations. NOTE—A <u>file</u> exists nominally in a directory, and can have an associated path.
intelligent transportation systems (ITS)	The application of advanced information processing and communications, sensing, and control technologies to surface transportation with the objective of promoting more efficient use of the existing highway and transportation network, increasing safety and mobility, and decreasing environmental impacts. NOTE—Also known as “intelligent transport systems”
International Organization for Standardization (ISO)	An international standards organization. NOTE—ANSI is the primary interface to ISO within the United States. Often thought to be International Standards Organization, because of its acronym ISO.
internet	A large collection of connected networks, primarily in the United States, running the Internet suite of protocols. NOTE—Sometimes referred to as the <i>DARPA Internet</i> , <i>NSF/DARPA Internet</i> , or the <i>Federal Research Internet</i> .
Interchangeable	A condition which exists when two or more items possess such functional and physical characteristics as to be equivalent in performance and durability, and are capable of being exchanged one for the other without alteration of the items themselves, or adjoining items, except for adjustment, and without selection for fit and performance. Note: See National Telecommunications and Information Administration, U.S. Department of Commerce
interoperability	Degree to which two or more systems, products or components can exchange information and use the information that has been exchanged.

Term	Definition
	[ISO/IEC/IEEE 24765]
leaf object type	An <u>object type</u> that does not have any sub-identifiers assigned (other than for <u>object</u> instances)
management information base (MIB)	A structured collection of managed information contained within a <u>SNMP context</u> . A MIB represents the instantiated data objects defined by one or more MIB modules and integrated into a device or management station.
Management Station	The computer system with which the device communicates. Typically, the management station commands and monitors the device.
MIB module	A sequence of <u>NVT ASCII</u> characters conforming to X.208 Abstract Syntax Notation One (<u>ASN.1</u>) and IAB STD 58. MIB modules are used to define the data objects that are to make up the instantiated objects of the MIB. NOTE—A <u>MIB module</u> can be, and within NTCIP typically is, presented as both part of a standard and as a stand-alone, computer-readable text file.
National Transportation Communications for ITS Protocol (NTCIP)	A family of protocols that provide common control and data collection services as well as accommodating various system topologies and data routing duties. NOTE—NTCIP is designed to support not only currently deployed systems, but also new systems and technologies as they become available.
network	A collection of devices connected by intermediate systems and populated by end systems.
notification	An unsolicited event message issued by an <u>SNMP agent</u> for a management station with an expectation of an acknowledgement.
object	An entity identified by a node on the international <u>object</u> identifier tree. NOTE – See <u>object instance</u> and <u>object type</u> , both of which are types of objects, along with any other entity that can be identified, such as a standard. This document attempts to use the most precise term within its text to assist in distinguishing among "object type" or "object instance". The use of the term " <u>object</u> " is typically limited to registering items on the international <u>object</u> identifier tree.
OBJECT IDENTIFIER	A unique name (identifier) that is associated with each type of <u>object</u> in a MIB that is a defined <u>ASN.1</u> type.
object instance	An instance of an <u>object type</u> , which is an actual instance of data. NOTE – This document avoids using the term " <u>object</u> " to describe this concept to prevent any ambiguity with an <u>object</u> referenced by the international <u>object</u> identifier tree or an <u>object type</u> .
object type	An abstract specification for a specific piece of data that can be instantiated within a device (zero or many times depending on how it is specified). It specifies the data using the <u>OBJECT-TYPE</u> macro.
OBJECT-TYPE	An <u>X.208 ASN.1</u> macro used to define the meta-attributes of an <u>object type</u> in an <u>SNMP MIB module</u> .
obsolete	A status value that indicates the definition is no longer valid, was found to be flawed, was redundant, or was not useful. NOTE—In the next (or some future) edition of a standard, any element (e.g., <u>requirement</u> , <u>object</u>) with a status of " <u>obsolete</u> " can be removed. This definition is modified from "Understanding <u>SNMP MIBS</u> ."
protocol	A specific set of rules, procedures, and conventions defining the format and timing of data transmissions between devices that are accepted and used to understand each other.
Requirement	A <u>requirement</u> describes a condition or capability to which a system shall conform; either derived directly from <u>user needs</u> , or stated in a contract, standard, specification, or other normative document. A desired feature, property, or behavior of a system.
Requirements Traceability	The ability to follow or study the logical progression among the needs, requirements, and design details in a step-by-step fashion.

Term	Definition
Simple Network Management Protocol (SNMP)	A communications protocol developed by the Internet Engineering Task Force (IETF), used for configuration and monitoring of network devices.
Trap	An unsolicited event message issued by an SNMP agent for a management station without any expectation of an acknowledgement.
User	A person using the system that is developed.
User Need	The business or operational problem (opportunity) that is to be fulfilled to justify procurement or use. Note: While this is termed a “user need” within the NTCIP community, it reflects needs of all stakeholders.

1.5 Abbreviations

The abbreviations and acronyms used in this document are defined as follows:

ASCII	American National Standard Code for Information Interchange
ASN.1	Abstract Syntax Notation One, as defined by either ITU-T X.208 or ITU-T X.680 (ISO/IEC 8824-1) (within the context used, both apply)
BER	Basic Encoding Rules, as defined by ITU-T X.690 (ISO/IEC 8825-1)
IAB STD	Internet Architecture Board Standard
IANA	Internet Assigned Numbers Authority
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IETF	Internet Engineering Task Force
IP	Internet Protocol
ITU-T	International Telecommunication Union – Telecommunication Standardization Sector
MVI	Multi-Version Interoperability (formerly Backward Compatibility)
NVT	Network Virtual Terminal
OER	Octet Encoding Rules, as defined by ITU-T X.696 (ISO/IEC 8825-7)
OSI	Open Systems Interconnection
PDU	Protocol Data Unit
PRL	Protocol Requirements List
RFC	Request for Comments
SMI	Structure and Identification of Management Information
SNMP	Simple Network Management Protocol version 3
TCP	Transport Control Protocol
UML	Unified Modeling Language
X.208 ASN.1	Abstract Syntax Notation One, as defined by ITU-T X.208
X.680 ASN.1	Abstract Syntax Notation One, as defined by ITU-T X.680 (ISO/IEC 8824-1)

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- ¹. [IAB STD 58](#) is still based on [ITU-T X.208](#), which is now withdrawn, but defines the fundamental concept of a macro and [MIB module](#) as used for [SNMP](#). ←
 - ². See the [ISO maintenance portal for ISO 26048-1](#) for maintained electronic files. ←

 June 21, 2025

Policy, Process and Procedure Management

Policy, Process and Procedure Development

It is the responsibility of the NTCIP JC to approve the policies, processes and procedures for the NTCIP Project. The NTCIP Coordinator reviews the policies, processes, and procedures with the NTCIP JC on a yearly basis to determine their effectiveness. The steps for establishing a new policy, process or procedure are as follows:

1. Any contributor to the project may make a proposal to the NTCIP JC Chairs or the NTCIP Coordinator that the policies, processes and procedures be updated to cover a matter concerning the NTCIP project.
2. The NTCIP JC Chairs and the NTCIP Coordinator discuss the proposal and determine if it is appropriate for consideration by the NTCIP JC.
3. If the NTCIP JC Chairs and the NTCIP Coordinator determine that the proposal is appropriate for NTCIP JC consideration, then the proposal is presented to the NTCIP JC for discussion; otherwise, the proposal is denied and the contributor is notified.
4. If it is the consensus of the NTCIP JC that the update to the policies, processes and procedures is to be developed, then a) a resource (or resource team) is identified to draft the policies, processes and procedures and b) a schedule is established for completion of the work; otherwise, the proposal is denied and the contributor is notified.
5. If a resource is not identified or a schedule is not established, then the effort does not proceed and the contributor is notified.
6. The modifications to the draft policies, processes and procedures are made and submitted to NTCIP JC for review.
7. If it is the consensus of the NTCIP JC that the draft policies, processes and procedures are acceptable, then the NTCIP Coordinator integrates the updated policies, processes and procedures into the NTCIP project as appropriate; otherwise,
 - a. The NTCIP JC provides comments for suggested changes and the development continues from Step 5 or
 - b. The work on the policies, processes and procedures concludes (does not continue) and the contributor is notified.

Minor corrections or additions to the policies, processes and procedures may be made by the NTCIP Coordinator or another contributor and approved by the Chair of the NTCIP JC. In such cases, the NTCIP JC is notified of the change. If a proposal for an update to the policies, processes and procedures does not go forward because a resource was not available or the work could not be completed in a timely manner, the NTCIP Coordinator may add the uncompleted work or a summary of the proposed change to Appendix A of this document for future consideration.

Policy, Process and Procedure Style

Policies, processes and procedures are to be written in the present tense. The word "shall" is to be avoided. Instead of using "shall" use: "must" for an obligation, "must not" for a prohibition, "may" for a discretionary action and "should" for a recommendation.

Section 3 Policies

This section contains the policies of the NTCIP project.

Each policy has the following outline:

- **Section Number and Title** – The title is not to be excessively long.
- **Effective Date** – Date when the policy took effect.
- **Approved By** – Typically the NTCIP Coordinator or the NTCIP JC.
- **Policy Contact** – The point of contact when the policy needs to change. The point of contact is to be one of the project rolls identified in Section 1.5.
- **Supersedes** – Title and date of any policies superseded by this policy.
- **Last Reviewed/Updated** – Date when the policy was last reviewed or updated.
- **Applies To** – The project rolls of those the policy applies to.
- **History** – History of major changes to the policy.
- **Related Policies** – Policies that have a related concern or are causally connected.
- **Policy Statement** – State the policy. Include: a) the situation(s) when this policy does or does not apply, b) the major conditions or restrictions and c) any exclusions or special situations.

Definitions of terms and acronyms used in the policies are included in Section 1.5 and Section 1.6, respectively.

3.1 NTCIP Consensus-Based Organization

Effective Date:	September 30, 2017
Approved By:	NTCIP Joint Committee
Policy Contact:	NTCIP Coordinator
Supersedes:	N/A
Last Reviewed/Updated:	September 30, 2017
Applies To:	NTCIP Joint Committee and Working Groups
History:	None
Related Policies:	None

3.1.1 Policy Statement

The AASHTO, ITE and NEMA develop and issue standards that are widely accepted and deployed in the transportation field, particularly as they apply to transportation infrastructure, traffic management and control procedures, and electronic traffic control equipment. In 1996, they established a MOU for the formation and operation of the NTCIP JC (see Section 1.4 References). The NTCIP JC is made up of 18 voting members with one-third (1/3) each being

designated by AASHTO, ITE and NEMA. Non-voting members of the NTCIP JC may include staff representatives from each of the SDOs (SDO Staff), liaison representatives from the USDOT and other liaison representatives as identified by the Chair of the NTCIP JC.

The purpose of the NTCIP project is to develop and maintain the NTCIP family of standards which are designed to achieve interoperability between transportation management centers and transportation field equipment and between transportation management centers and central systems of various types.

To accomplish this, the NTCIP project employs a consensus-based organizational structure that provides for layers of stakeholder review, comment collection and resolution, and acceptance (see Figure 1.).

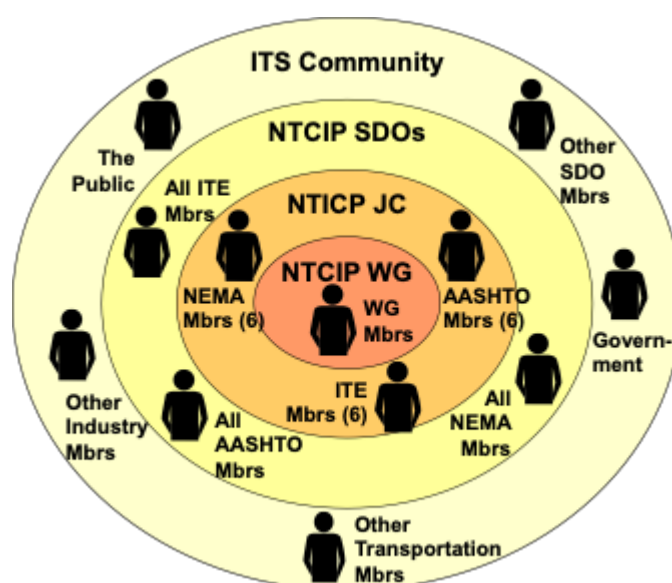


Figure 1: Consensus-based organization for the development of NTCIP standards

Central to the NTCIP consensus organization are the Working Groups which function under the oversight of the NTCIP JC to perform the technical work of developing and maintaining NTCIP standards and related documents. These WGs are created and tasked with work items by the NTCIP JC. WGs may be made up of consultants, manufacturers, software providers, public transportation professionals, users, and even those in industries not traditionally considered transportation. Membership in one of the NTCIP SDOs is not a requirement to be a member in a WG. The NTCIP JC oversees the activities of the WGs, reviews work items produced by the WG, and determines if the work items are suitable for distribution through the memberships of the NTCIP SDOs. The SDOs of the NTCIP project still retain their individual internal standards approval (or adoption) processes. Once all three NTCIP SDOs have approved an NTCIP standard,

the standard is said to be “Jointly Approved.” The ITS Community represents those who are largely beneficiaries of NTCIP standards who are not already included in the other layers.

3.2 NTCIP Document Classifications

Effective Date	September 30, 2017
Approved By	NTCIP Joint Committee
Policy Contact	NTCIP Coordinator
Supersedes	N/A
Last Reviewed/Updated	September 30, 2017
Applies To	NTCIP Joint Committee and Working Groups
History	None
Related Policies	None

3.2.1 Policy Statement

There are a variety of document types covered by the NTCIP efforts, including:

1. Standards;
2. Errata;
3. Process, Control, and Information Management Policies; and
4. Informational Reports.

3.2.2 Standards

The primary activity of the NTCIP project effort is to develop standards. NTCIP Standards include those documents that define specific requirements related to protocols, data definitions, message set definitions, etc. and that are developed and maintained in a committee environment. Standards are divided into Base Standards and Profiles.

The scheme for document numbering and the procedures for configuration management are defined in NTCIP 8002.

3.2.2.1 Base Standards

Base standards are those standards that define the details of each specific feature. The features themselves may be optional or mandatory. There are three major categories of base standards as defined below.

3.2.2.2.0 PROTOCOLS

Protocols define the basic requirements of the various NTCIP protocols and procedures.

3.2.2.3.0 DATA DICTIONARIES

Data Dictionaries, sometimes called *Object Definitions*, define the exact meaning of specific data elements.

3.2.2.4.0 MESSAGE SETS

Message sets define the exact structure and meaning of specific messages or groups of data elements.

3.2.2.5 Profiles

Profiles are standards that reference one or more other standards to define the details of implementation. In general, profiles are used to define how to combine a set of base standards to achieve a specific set of functionality. The profiles will often require the implementation of certain optional features of base standards to ensure interoperability of the desired features. Profiles are useful as they provide an unambiguous definition of how to implement base standards.

See NTCIP 8002 for the definitions of the types, or levels, of profiles, such as Information Profiles, Application Profiles, Transport Profiles, and Subnet Profiles.

3.2.3 Errata Document

An errata document may be developed and published to correct technical issues, clarify ambiguities, or correct editorial mistakes in a published standard. It is not intended to add new features or change the intent of the features in the published standard. An errata document developed for a standard is not required to go through a user comment period or through an

SDO ballot or approval. However, the NTCIP JC may require a user comment period if the corrective measures in the errata are extensive or complicated.

3.2.4 Process, Control, and Information Management Policies

Another series of NTCIP publications are those documents defining how the NTCIP committees conduct business and produce reports, such as this document. These documents are categorized as Process, Control, and Information Management Policies.

3.2.5 Informational Reports

In several cases, the NTCIP effort has identified a need to provide informative documents to better explain certain aspects of the NTCIP. These documents can cover a wide variety of topics from detailed explanations of how to use certain features of the protocol to reports identifying where the NTCIP has been used. The documents can either be produced by the NTCIP effort itself, or may be submitted by any interested party for consideration by the NTCIP effort.

Documents that have undergone a rigorous review and consensus building process may be formally recognized by the NTCIP process and assigned an NTCIP number for configuration management purposes. The Joint Committee may also decide to post other informational materials on the NTCIP Web Site if it deems appropriate.

Section 4 Processes

This section contains the processes of the NTCIP project.

Each process has the following outline:

- Section Number and Title – The title is not to be excessively long.
- Effective Date – Date when the process took effect.
- Approved By – Typically the NTCIP Coordinator or the NTCIP JC.
- Process Contact – The point of contact when the process needs to change. The point of contact is to be one of the project rolls identified in Section 1.5.
- Supersedes – Title and date of any processes superseded by this process.
- Last Reviewed/Updated – Date when the process was last reviewed or updated.
- Applies To – The project rolls of those the process applies to.
- History – History of major changes to the process.
- Related Processes – Processes that have a related concern or are causally connected.
- Process Statement – State the process.

Definitions of terms and acronyms used in the policies are included in Section 1.5 and Section 1.6, respectively.

4.1 NTCIP Standards Development Process

Effective Date: September 30, 2017

Approved By: NTICP Joint Committee

Process Contact: NTCIP Coordinator

Supersedes: N/A

Last Reviewed/Updated: September 30, 2017

Applies To: NTCIP Joint Committee and Working Groups

History: None

Related Processes: None

Process Statement:

In concept, the process of creating a standard is straightforward: a specification undergoes a period of development and several iterations of review and revision, is approved (or adopted) as a standard by the appropriate body, and is published. In practice, the process is more complicated, due to:

1. the difficulty of creating specifications of high technical quality;
2. the need to consider the interests of all of the affected parties;
3. the importance of establishing widespread community consensus; and
4. the difficulty of evaluating the utility of a particular specification.

The goals of the NTCIP Standards Process are:

1. technical excellence;
2. clear, concise, and easily understood documentation;
3. openness and fairness; and
4. meeting needs within the ITS community.

The goals of technical excellence and the need to allow all interested parties to comment on work items require significant time and effort. On the other hand, today's rapid development of technology demands the timely development of standards. The NTCIP standards process is intended to balance these conflicting goals. The process is believed to be as short and simple as possible without unduly sacrificing technical excellence or openness and fairness.

From its inception, the NTCIP project has been, and is expected to remain, an evolving family of standards whose participants regularly factor new requirements and technology into its design and implementation. Users of NTCIP standards and providers of the equipment, software, and services that support it should anticipate and embrace this evolution as a major tenet of NTCIP philosophy.

The NTCIP standards development process is shown in Figure 2. It includes steps of increasing levels of review and endorsement. Within each level, there are also several steps related to specific activities that take place.

4.1.1 Step 1 – JC Assigns a Standards Work Item to a WG

A standards activity begins with a proposal submitted to the NTCIP JC for consideration (see Procedure 5.2 NTCIP JC Proposals). Typically, a proposal comes from an existing WG but it may come from other sources such as a member of the NTCIP JC, an SDO, or the USDOT. If the JC chooses to go forward with the proposal, a work item is identified (see Policy 3.2 NTCIP Document Classifications) and assigned to an appropriate WG. If no appropriate WG exists, a new working group may be formed (see Procedure 5.1.1 Creating an NTCIP Working Group).

Once the work item is assigned to a WG and any preconditions for the proposal are met, then the proposal is approved and work may begin.

4.1.2 Step 2 – WG Prepares Working Group Draft (WGD) Standards

After a proposal has been approved, the WG begins the task of creating draft standards for internal use by the WG. Typically, a paid consultant or consulting team performs this work. However, the work may also be done on a volunteer basis by WG members.

Early WGD standards may not be complete, may only address certain features or areas of the standard, and may not be in an NTCIP standard format (see NTCIP 8002). As the WG continues with its development, a WGD standard will take on the NTCIP standard format and the necessary sections of the standard are drafted.

4.1.3 Step 3 – WG Prepares Proposed User Comment Draft (pUCD) Standard

Once a WGD standard is substantially complete, the WG may take a vote to advance the WGD standard to the NTCIP JC as a pUCD standard. To qualify as a pUCD standard, the standard must adhere to NTCIP 8002 and have all major technical issues addressed. If a vote on a pUCD standard fails, the WG may continue to work on the standard and votes retaken until pUCD standard is achieved.

4.1.4 Step 4 – JC Reviews pUCD

The pUCD standard is distributed to the NTCIP JC for review by the members. Members should formulate comments and questions prior to a NTCIP JC meeting with the WG Chair.

4.1.5 Step 5 – JC Accepts pUCD as User Comment Draft (UCD) Standard

A meeting, web conference or teleconference is held where the WG Chair (and possibly a technical lead in the project) presents the pUCD standard to the members of the NTCIP JC. The NTCIP JC asks the presentation team questions and makes comments based on their review. Any issue may be discussed regarding the pUCD standard. The NTCIP JC may take a vote to advance the pUCD as a User Comment Draft (UCD) standard. To qualify as UCD standard, the document must relate to an approved proposal and submitted by the authorized WG. If a vote on a UCD standard fails, the NTCIP will provide direction on how to proceed. This direction may be to simply correct editorial items in the standard prior to a revote or there may be some more

significant development to be done. When the NTCIP JC votes to accept the pUCD as a UCD, it is advanced to the SDOs.

4.1.6 Step 6 – SDOs Distribute UCD for Review and Solicits Comments

The NTCIP SDOs distribute the UCD standard to the members of their organizations and the UCD standard is posted on the NTCIP Web Site. A user comment period is identified and comments on the UCD standard are solicited. A user comment period must not be less than 30 calendar days.

4.1.7 Step 7 – WG Collects and Resolves Comments

During the user comment period, the WG collects all comments received whether the source of a comment is from a user or otherwise. Comments may be sent directly to the WG or forwarded through SDO Staff to the WG. The comments may be editorial, technical, simple or substantial in nature.

The WG adjudicates each comment submitted during the user comment period. A comment may be “rejected” if it is the consensus of the WG that the comment was not proper or appropriate. Some of the reasons that a comment may be rejected are that the WG may disagree technically with the proposed change, the WG may decide that the proposed change is out of scope for the standard being developed, or the comment may not be suitable because of other corrective actions taken by the WG. A comment is “accepted” when it is the consensus of the WG that the suggested change or issue is appropriate. Often, comments submitted include suggested changes to the text of the standard. Such changes are taken as recommendations but the WG may accept the comment but modify the corrective measures suggested. The adjudications of the comments are posted on the NTCIP Web Site, sent back to the commenters or both.

4.1.8 Step 8 – WG Prepares Proposed Recommended Standard (pRS)

Based on the adjudication of comments from the user comment period, the WG updates the standard accordingly. Once the standard is substantially complete, the WG may take a vote to

advance the standard to the NTCIP JC as a pRS. If a vote on a pRS standard fails, the WG may continue to work on the standard and votes retaken until pRS is achieved.

4.1.9 Step 9 – JC Reviews pRS

The pRS is distributed to the NTCIP JC for review by the members. Members should formulate comments and questions prior to a NTCIP JC meeting with the WG Chair.

4.1.10 Step 10 – JC Accepts pRS as RS

A meeting, web conference or teleconference is held where the WG Chair (and possibly a technical lead in the project) presents the pRS to the members of the NTCIP JC. The NTCIP JC asks the presentation team questions and makes comments based on their review. Any issue may be discussed regarding the pRS. The NTCIP JC may take a vote to advance the pRS as a Recommended Standard (RS). To qualify as an RS, the document must relate to an approved proposal and submitted by the authorized WG. If a vote on an RS fails, the NTCIP JC will provide direction on how to proceed. This direction may be to simply correct items in the standard prior to a revote or there may be some more significant development to be done. When the NTCIP JC votes to accept the pRS as an RS, it is advanced to the SDOs.

4.1.11 Step 11 – SDOs Distribute RS for Review

The NTCIP SDOs distribute the RS to the members of their organizations and the RS standard is posted on the NTCIP Web Site. The RS review period must not be less than 30 calendar days. This review is part of the approval (or ballot) process.

4.1.12 Step 12 – NTCIP SDOs Approve Standard

Each SDO has their own approval (or ballot) process used within their organization. The SDOs may allow “ballot comments” to be submitted during the review that must be adjudicated by the WG and their resolution approved by the NTCIP JC. The SDOs may provide an opportunity for individuals to make an appeal against the approval of a standard. Resolution of an appeal may also require actions from the WG or NTCIP JC. Once all three NTCIP SDOs approve (or adopt) the standard, the standard is considered a Joint Standard of AASHTO, ITE and NEMA.

4.1.13 Step 13 – Publish Jointly Approved Standard

The standard is updated with any final editorial corrections. It is posted on the NTCIP Web Site by the NTCIP Coordinator.



Figure 2: Standards development process

4.2 NTCIP Errata Development Process

Effective Date	September 30, 2017
Approved By	NTICP Joint Committee
Process Contact	NTCIP Coordinator
Supersedes	N/A
Last Reviewed/Updated	September 30, 2017
Applies To	NTCIP Joint Committee and Working Groups
History	None
Related Processes	None

4.2.1 Process Statement

Once a standard is developed and published, deployments of the standard in devices and systems will uncover errors and ambiguities in the standard not foreseen prior to publication. In

order to correct such issues and facilitate future deployments of the standard, an errata document may be developed and published. An abbreviated process compared to that of the standards development process is used. An errata document developed for a standard is not required to go through a user comment period or through an SDO ballot or approval. However, the NTCIP JC may require a user comment period if the corrective measures in the errata are extensive or complicated. See Figure 3.

4.2.2 Step 1 – JC Assigns an Errata Work Item to a WG

A standards activity begins with a proposal submitted to the NTCIP JC for consideration (see Procedure 5.1.2 NTCIP JC Proposals). Typically, a proposal comes from an existing WG but it may come from other sources such as a member of the NTCIP JC, an SDO, or the USDOT. If the JC chooses to go forward with the proposal, a work item is identified (see Policy 3.2 NTCIP Document Classifications) and assigned to an appropriate WG. Once the work item is assigned to a WG and any preconditions for the proposal are met, then the proposal is approved and work may begin.

4.2.3 Step 2 – WG Prepares Working Group Draft (WGD) Errata

After a proposal has been approved, the WG begins the task of creating draft errata for internal use by the WG. Typically, a paid consultant or consulting team performs this work. However, the work may also be done on a volunteer basis by WG members.

4.2.4 Step 3 – WG Prepares Proposed Recommended Errata (pRE)

Once the errata document is substantially complete, the WG may take a vote to advance the errata to the NTCIP JC as a pRE. If a vote on a pRE document fails, the WG may continue to work on the errata and votes retaken until a pRE errata is achieved.

4.2.5 Step 4 – JC Reviews pRE

The pRE is distributed to the NTCIP JC for review by the members. Members should formulate comments and questions prior to a NTCIP JC meeting with the WG Chair.

4.2.6 Step 5 – JC Accepts pRE as an RE

A meeting, web conference or teleconference is held where the WG Chair (and possibly a technical lead in the project) presents the pRE to the members of the NTCIP JC. The NTCIP JC asks the presentation team questions and makes comments based on their review. Any issue may be discussed regarding the pRE. The NTCIP JC may take a vote to advance the pRE as a

Recommended Errata (RE). If a vote on an RE fails, the NTCIP JC will provide direction on how to proceed. This direction may be to simply correct items in the errata prior to a revote or there may be some more significant development to be done. When the NTCIP JC votes to accept the pRE as an RE, it is qualified for publication.

4.2.7 Step 6 – Publish Jointly Approved Errata

The errata document is updated with any final editorial corrections. It is posted on the NTCIP Web Site by the NTCIP Coordinator.

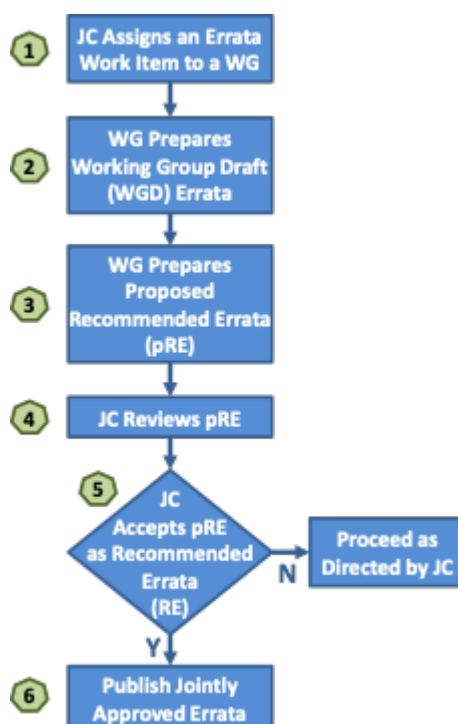


Figure 3: Errata development process

Section 5 Procedures

This section contains the procedures of the NTCIP project.

Each procedure has the following outline:

- Section Number and Title – The title is not to be excessively long.
- Effective Date – Date when the procedure took effect.
- Approved By – Typically the NTCIP Coordinator or the NTCIP JC.
- Procedure Contact – The point of contact when the procedure needs to change. The point of contact is to be one of the project rolls identified in Section 1.5.
- Supersedes – Title and date of any procedures superseded by this procedure.
- Last Reviewed/Updated – Date when the procedure was last reviewed or updated.
- Applies To – The project rolls of those the procedure applies to.
- History – History of major changes to the procedures.
- Related Procedures – Procedures that have a related concern or are causally connected.
- Procedure Statement – State the procedure.

Definitions of terms and acronyms used in the policies are included in Section 1.5 and Section 1.6, respectively.

5.1 Procedures of the NTCIP JC

This section contains the procedures that primarily address activities of the NTCIP JC.

5.1.1 Creating an NTCIP Working Group

Effective Date	September 30, 2017
Approved By	NTCIP Joint Committee
Procedure Contact	NTCIP Coordinator
Supersedes	N/A
Last Reviewed/Updated	September 30, 2017
Applies To	NTCIP Joint Committee and Working Groups
History	None
Related Procedures	5.2.1 NTCIP WG Organization

5.1.2 Procedure Statement

There may be times when work item to be undertaken by the NTCIP JC is not suitable for an existing NTCIP WG. In this case, the NTCIP JC may create a new WG as follows:

1. The need for the new group must be stated and recorded.
2. The work items to be assigned to the group identified.
3. A NTCIP JC voting member identified to act as a sponsor help support the WG in its activities;
4. The NTCIP to assign a Chair or Chairs of the WG with input from the NTCIP JC, the SDO representatives and the NTCIP Coordinator.
5. The initial voting membership of the WG is proposed by the Chair or Chairs until it is suitable to the NTCIP JC.
6. Confirmation of this is to be done via a motion and pass via a super majority of two-thirds (2/3s) or more affirmative votes from of the current voting members of the NTCIP JC. For example, if there are 17 voting members, then two-thirds is 11.33 which means there are 12 affirmative votes or more are required.

5.1.3 NTCIP JC Proposals

Effective Date	September 30, 2017
Approved By	NTCIP Joint Committee
Procedure Contact	NTCIP Coordinator
Supersedes	N/A
Last Reviewed/Updated	September 30, 2017
Applies To	NTCIP Joint Committee and Working Groups
History	None
Related Procedures	None

5.1.3.1 Procedure Statement

All documents developed under the oversight of the NTCIP JC begin with a proposal (Registries excepted, see Policy 3.2 NTCIP Document Classifications). When a proposal is submitted to the Joint Committee, it will include the following details regarding the effort:

1. the technical lead for the document to be developed;
2. the working group, or other team, responsible for assisting the technical lead;
3. the scope of the document to be developed;
4. an explanation of why the document is needed;
5. any related documents;
6. any related working groups;
7. the expected completion date; and
8. an estimate of any required financial assistance.

A proposal may also contain sample text for the document. The sample text can be anything from a few words to an entire draft.

Proposals are submitted to the NTCIP Joint Committee for approval; however, most proposals should be submitted through an appropriate working group in order to show a level of consensus for the proposal.

5.1.4 NTCIP JC Vacancies

Effective Date	September 30, 2017
Approved By	NTCIP Joint Committee
Procedure Contact	NTCIP Coordinator
Supersedes	N/A
Last Reviewed/ Updated	September 30, 2017
Applies To	NTCIP Joint Committee and Working Groups
History	This procedure on NTCIP JC Vacancies was established to provide the NTCIP JC the means to fill its own vacancies on the NTCIP JC if the SDOs are unable to do so.
Related Procedures	None

5.1.4.1 Procedure Statement

The procedures for membership and leadership and of the NTCIP JC are established in the MOU Section 1.0 Establishment of a Joint Committee. The MOU states that there are “18 voting members with 1/3rd each being designated by AASHTO, ITE, and NEMA.” While this is the intent, it is often the case that that an SDO is not able to immediately fill a vacancy when one occurs. In

order for the NTCIP JC to be effective, its voting membership must be maintained. The steps below help insure the NTCIP JC membership.

1. A vacancy on the NTCIP JC may identified by an SDO or the NTCIP Chair when they are informed by a voting member that they will no longer be participating in that capacity. A vacancy may also be identified by the NTCIP Chair that a voting member has not participated in three consecutive NTCIP JC meetings over a three month period or greater.
2. The SDO which has the vacancy on the NTCIP JC is allowed up to one year to fill a vacancy.
3. If the SDO does not fill the vacancy in one year, the NTCIP JC Chair may nominate a representative pro tem to fill the vacancy. The representative pro tem must be a member (or representative of a member organization) of the SDO that has the vacancy on the NTCIP JC. The NTCIP JC must confirm the nomination by simple majority.

5.1.5 Meeting Procedures of the NTCIP JC

Effective Date	September 30, 2017
Approved By	NTICP Joint Committee
Procedure Contact	NTCIP Coordinator
Supersedes	N/A
Last Reviewed/ Updated	June 21, 2018
Applies To	NTCIP Joint Committee and Working Groups
History	6/21/2018 Decreased the number of votes required for Recommended Standards and User Comment Drafts. 9/30/2017 This procedure on meeting Requirements of the NTCIP JC changes the previous practices in NTCIP 8001 v01.08 that assumed a full 18 voting member NTCIP Joint Committee to one that is based on the current number of voting members. Made change to <u>requirement</u> of a simple majority.
Related Procedures	None

5.1.5.1 Procedure Statement

Much of the work of the NTCIP JC occurs in meetings. Meetings may take place in a face-to-face setting, through web conferencing or through a teleconference. The practices for conducting meetings of the NTCIP JC are described below.

Parliamentary Authority: The parliamentary authority of the NTCIP JC follows Robert's Rules of Order Newly Revised (RONR). RONR governs in all cases to which they are applicable and in which they are not inconsistent with this policies, processes and procedures document.

Quorum: The number of members required to hold a meeting of the NTCIP JC is one-third (1/3) or more of the current NTCIP JC voting membership with at least one voting member present from each of the NTCIP SDOs. For example, if there are 17 voting members, then one-third is 5.67 which means that there are 6 voting members present or more required.

Affirmative Votes for a Recommended Standard: This is the number of votes required for the NTCIP JC to accept a pRS as an RS or to accept a pRE as an RE. A recommended majority is two-thirds ($2/3$ s) or more affirmative votes at a meeting where a quorum is present. For example, if there are 17 voting members and 10 voting members present at a meeting, then two-thirds is 6.67 which means there are 7 affirmative votes or more required.

Affirmative Votes for a User Comment Draft: This is the number of votes required for the NTCIP JC to accept a pUCD as an UCD. A user comment majority is one-half plus one ($1/2 + 1$) or more affirmative votes at a meeting where a quorum is present. For example, if there are 17 voting members and 10 voting members present at a meeting, then one-half is 5 which means that there are 6 affirmative votes or more required.

Votes for Simple Majority: This is number of votes required for general decisions by the NTCIP JC not otherwise covered in this section. A simple majority vote is more than half the votes cast. A quorum must be present. Abstentions are not counted. For example, if there are six voting members present at meeting and the quorum requirement is met, then a vote of two yeas, one nay and three abstentions results in a motion passing. In the same meeting, a vote of three yeas and three nays results in a motion failing because a majority was not achieved.

Acceptance Actions Between Meetings of the NTCIP JC: Acceptance actions by the NTCIP JC are those when a vote is taken to accept a pUCD as a UCD, to accept a pRS as an RS, or to accept a pRE as an RE. Typically, acceptance actions are completed as part of an NTCIP JC meeting. There are circumstances, however, when the NTCIP JC may determine that it is prudent to complete an acceptance action outside of a meeting. This may be done because the NTCIP JC members need more time to review a proposed document before voting, there are not

enough voting members at the meeting to perform the vote, or some other reason. In such cases, an electronic ballot may be used outside of the NTCIP JC meeting as follows:

1. A motion must be made that an electronic ballot be taken to accept the pUCD, pRS or pRE as a UCD, RS or RE, respectively. The motion must be recorded at the meeting and include the details of when and how the ballot will be performed. Electronic balloting may include email, a polling website, or some other method.
2. The motion must be seconded.
3. The motion must pass by simple majority.
4. No split vote between a meeting and electronic voting is allowed. All voting must take place during a meeting NTCIP JC or electronically outside of the meeting but not both.

5.2 Procedures of the NTCIP WGs

This section contains the procedures that primarily address activities of the NTCIP WGs.

5.2.1 NTCIP WG Organization

Effective Date	September 30, 2017
Approved By	NTCIP Joint Committee
Procedure Contact	NTCIP Coordinator
Supersedes	N/A
Last Reviewed/ Updated	September 30, 2017
Applies To	NTCIP Joint Committee and Working Groups
History	This procedure on meeting NTCIP WG Organization changes abilities on observing members. NTCIP Working Group Bylaws in NTCIP 8001 v01.08 so that observing members may be assigned tasks at the WG Chair's discretion.
Related Procedures	5.1.1 Creating an NTCIP Working Group

5.2.1.1 Procedure Statement

This procedure describes the organization and membership of NTCIP WGs. There are three categories of members within NTCIP WGs: Voting members, observing members, and liaison members. Individuals or companies wishing to serve on an NTCIP WG must petition the Chair (or Chairs) for membership. The term of membership is three (3) years from the date of appointment. Members may be reappointed to additional terms.

WG Chair or Chairs: NTCIP WG Chairs are appointed by the NTCIP JC Chair for a term of three (3) years (see 5.1.1 Creating an NTCIP Working Group). A WG Chair is an “at will” appointment and can be terminated by either party within the term. WG Chairs may be reappointed to additional terms. A WG Chair is a voting member of the WG.

Voting Members: There are at least seven (7) and no more than nineteen (19) voting members. The voting members are eligible to vote on all business before the WG. There can be only one voting member from each organizational entity. An organizational entity may be a department of transportation, corporation, or association. The voting members are expected to actively participate in the meetings of the WG, and perform their assigned tasks. The voting members are appointed by the WG Chair. The initial roster of voting members must be approved by the NTCIP JC. Subsequent additions or cancellations of voting memberships are proposed by the WG Chair and approved by the NTCIP JC Chair. If a voting member has not attended or been represented at three (3) of the last four (4) regular meetings, their membership category may be changed to observing member. In such a case, the member is to be notified by the WG Chair. A voting member may designate an alternate voting member from the same organizational entity to represent the voting member at a WG meeting. The alternate voting member is eligible to vote on all business before the WG.

Observing Members: There are no limits on the number of observing members. The observing members are those who choose to participate at various levels of regularity but are not voting members. Observing members are appointed by the WG Chair. Observing members may be assigned tasks at the WG Chairs discretion. An observing member may designate an alternate observing member from the same organizational entity to attend meetings in their absence.

Liaison Members: Liaison members are the employees of standards developing organization and sponsoring agencies in the federal government who, in their official capacity, would like to be informed on the activities of the WG. Liaison members are non-voting members. Liaison

members are appointed by the WG Chair. A liaison member may designate an alternate liaison member from the same organizational entity to attend meetings in their absence.

5.2.2 Meeting Procedures of NTCIP WGs

Effective Date	September 30, 2017
Approved By	NTCIP Joint Committee
Procedure Contact	NTCIP Coordinator
Supersedes	N/A
Last Reviewed/Updated	September 30, 2017
Applies To	NTCIP Joint Committee and Working Groups
History	None
Related Procedures	None

5.2.2.1 Procedure Statement

Much of the work of NTCIP WGs occur in meetings. Meetings may take place in a face-to-face setting, through web conferencing or through a teleconference. The practices for conducting meetings of the NTCIP WGs are described below.

Parliamentary Authority: The parliamentary authority of an NTCIP WG follows Robert's Rules of Order Newly Revised (RONR). RONR governs in all cases to which they are applicable and in which they are not inconsistent with this policies, processes and procedures document.

Quorum: The number of members required to hold a meeting of an NTCIP WG is one-half (1/2) or more of the current NTCIP WG voting membership. Alternate voting members present at a meeting are counted as voting members if the voting member is not present. For example, if there are 9 voting members, then one-half is 4.5 which means that there are 5 voting members present or more required.

Votes for Simple Majority: This is number of votes required for general decisions by the NTCIP WG not otherwise covered in this section. A simple majority vote is more than half the votes cast. A quorum must be present. Abstentions are not counted. For example, if there are six voting members present at meeting and the quorum requirement is met, then a vote of two yeas, one nay and three abstentions results in a motion passing. In the same meeting, a vote of three yeas and three nays results in a motion failing because a majority was not achieved.

Acceptance Actions Between Meetings of the NTCIP WG: Acceptance actions by the NTCIP WG are those when a vote is taken to accept a WGD as a pUCD, pRS or pRE. Typically, acceptance

actions are completed as part of an NTCIP WG meeting. There are circumstances, however, when the NTCIP WG may determine that it is prudent to complete an acceptance action outside of a meeting. This may be done because the NTCIP WG members need more time to review a proposed document before voting, there are not enough voting members at the meeting to perform the vote, or some other reason. In such cases, an electronic ballot may be used outside of the NTCIP WG meeting as follows:

1. A motion must be made that an electronic ballot be taken to accept the pUCD, pRS or pRE. The motion must be recorded at the meeting and include the details of when and how the ballot will be performed. Electronic balloting may include email, a polling website, or some other method.
2. The motion must be seconded.
3. The motion must pass by simple majority.
4. No split vote between a meeting and electronic voting is allowed. All voting must take place during a meeting NTCIP WG or electronically outside of the meeting but not both.

 June 21, 2025

Annex A PROPOSED UPDATES TO NTCIP 8001

1. Should there be observing members for the JC? How do these differ from observing members for WGs? Maybe we should have memberships as follows: voting, non-voting, observational and liaison. In this case, observational would not be given tasks while voting and non-voting members would. Non-voting members would be those that attend regularly but do not have voting privileges.
2. Procedures for reaffirming and rescinding documents should be added in. Note that amended and revised documents have been dropped.

Re-affirmed

NTCIP Standards expire after a period of five years unless they are revised or re-affirmed. In order to be re-affirmed, the exact same text of the standard (without any changes) must be re-balloted by the SDOs and approved. Thus, re-affirmed standards have demonstrated a great deal of stability.

Rescinded

If an NTCIP standard is no longer needed, it can be rescinded, either by vote or by allowing the five-year cycle to lapse. A status of rescinded indicates that there is no longer any version of this standard that is recommended for use.

1. Possibly add duties of the WG Chairs to existing procedures. Duties of the NTCIP Chair covered in MOU?
2. Review all of the headers of the policies, processes and procedures. Update as appropriate. Add salient points to history.
3. There are constraints on term for WG Chairs that have not been enforced in the past. Do we want to keep them in NTCIP 8001?
4. Review 8001_Annex_D WG Bylaws v06 r0812 to see if something was missed. Example: A former voting member getting restored to voting member status.
5. Need to add a process for approving a document that falls under the Process, Control, and Information Management Policies.

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